

IN BRIEF

The brain has evolved to withstand perturbations of every kind and yet, perturbation is also neuroscience's main tool for understanding it. We broadly accept that *there is no causation without manipulation*, but we understand surprisingly little about what those manipulations actually do to the brain, or how the brain counters them. Without this mechanistic understanding, the downstream effects of an intervention, be it a stroke or a stimulation protocol, remain hard to predict or to interpret. My research programme sets out to uncover **the principles of brain resilience** across multiple scales of organization, the human lifespan, and different species. I combine game theory, network science, dynamical systems theory, and brain-inspired AI models to build rigorous frameworks for causal inference in neural systems and use these tools to understand what perturbations do to the brain. In particular, I study how the brain's wiring and biophysical properties shape its communication dynamics, and which disruptions to this neuronal orchestra give rise to neuropsychiatric disorders. My aim is to build reliable maps from disruptions in brain networks to their neuropsychiatric consequences so we can characterize the effects of interventions and design better ones.

CURRENT POSITIONS

2026 – : Chris Rokos Fellow, Queens' College, University of Cambridge, UK

2025 – : Post-Doctoral Research Associate, 4D Lab, MRC Cognition and Brain Sciences Unit, University of Cambridge, UK

2025 – : Visiting Researcher, Institute of Computational Neuroscience (ICNS), University Medical Center of Hamburg (UKE), Germany

RESEARCH PROGRAMME AND OUTPUT

Aristotle provides a taxonomy for causes that I believe is still relevant for modern neuroscience. On one side of the spectrum lie **ultimate causes**, which asks teleological questions about *why* the effect should occur. On the other side sit **proximate causes**, which focuses on the immediate mechanisms underlying an effect. To address the first, I build normative models incorporating evolutionary and developmental constraints to understand what role those constraints play in shaping the brain's organization. To answer the second, my research operates on the principle that *"the functions a neural element undertakes are best established through systematic manipulations."*:

■ **Survival of the good-enough and the search for ultimate causes:**

Why is the brain organized the way it is? Ramon y Cajal's take, still widely accepted, was that neurons must carefully balance their wiring to achieve cost-efficient communication. Yet, it remained to be properly tested whether communication in the human brain is efficient at all and whether efficient communication is indeed *the* fundamental evolutionary pressure. I developed normative (game-theoretic) frameworks to address these questions (*eLife*, 2025, and *in press*, *Science Advances*). I found the answer to both to be "not really" as I show that human cortical networks trade both the energetic cost and fast communication for functional reliability, suggesting that **resilience** is instead a more influential architect of the brain's wiring diagram. In related work with collaborators, we formalized a well-established wiring principle, the avoidance of strong loops in the mammalian cortex, and show that brains across species could all have even a larger computational capacity by further reducing reciprocal connectivity (*Chaos*, 2026, and *under review*, *Nature Communications*). Lastly, I synthesized these findings into an opinion piece (*Trends in Cognitive Sciences*, 2026), arguing that the brain's apparent sub-optimality is by itself a great source of insight for both neuro-AI research and understanding brain function.

■ **Outmanoeuvring the brain and mapping out proximate causes:**

Brain-wide manipulations are yet to be widely adopted, but how to make sense of them when they are? To answer this, I validate my methods using virtual experiments on artificial neural networks (ANNs) and biophysical models of large-scale brain dynamics (BNNs), where every element can be lesioned and every consequence precisely measured. Collectively, my research showed that a unit's causal contribution can come apart from both its activity profile (*Scientific Reports*, 2024) and from its structural connectivity profile (*IEEE-IJCNN*, 2022), forcing us to ask the hard question: How can we then uncover the *true* causal relationships? I used a game-theoretic formalism built on **multi-site systematic perturbations** to axiomatically quantify those contributions in brain-inspired ANNs, hand-crafted AI models (*PLOS Computational Biology*, 2022, and *under review*, *npj Artificial Intelligence*), and BNNs (*eLife*, 2025). Together, these findings suggest that, unlike in engineered AI systems, brain networks are imbued with redundancy that make *causal inference through manipulations* far less trivial than we might have hoped.

HELPING THE NEXT GENERATION OF RESEARCHERS

In the age of AI and rapidly evolving neuroscience, supporting the next group of researchers is where I find a lot of joy in. I have taken on supervision and teaching whenever I had the opportunity to do so. For instance, I have supervised six masters, MPhil, and Erasmus projects across Cambridge, TU München, and Hamburg, on questions ranging from the brain's trade-off landscape to game-theoretic analysis of AI models and analytical models of inter-areal communication. I currently lecture on network neuroscience and computational modeling to MPhil students at the MRC-CBU, teach as a teaching assistant for Mathematical Biology at Pembroke College, and mentor projects at Neuromatch Academy, where I have contributed to since 2020.

Supervision

- 2026– MPhil thesis, "Inter-areal communication dynamics of the human brain across development", University of Cambridge.
- 2025–26 MSc thesis, "Mapping the trade-off landscape of the mammalian brain", University of Cambridge and Technische Universität München.
- 2022–24 MSc thesis, "Multidimensional game-theoretic analysis of AI models", ICNS-UKE.
- 2024 Two Erasmus projects, "Analytical models of inter-areal communication in brain networks", ICNS-UKE.

- 2023 MSc thesis, "Sub-optimal signal propagation in the human cortex", University of Sorbonne and ICNS-UKE.

Teaching

- 2026– Network Neuroscience lectures for MPhils, MRC-CBU, Cambridge.
- 2025– Teaching assistant for Mathematical Biology, Pembroke College, Cambridge.
- 2024– Project Mentor, Neuromatch Academy (online).
- 2020 Teaching assistant for Computational Neuroscience, Neuromatch Academy (online).

FROM INTER-AREAL BRAIN NETWORKS TO INTER-PERSONAL SOCIAL NETWORKS

Besides studying brain networks, I believe in the power of networks to connect people and make a difference. Because of that, I am editing a special issue that brings philosophers and neuroscientists together, co-organizing the Making Connections seminar series where we invite network scientists of all fields, and coordinating the Cambridge Networks Network, where we gather to foster interdisciplinary collaborations among network researchers. I am a founding member of the Bernstein society node in Hamburg and I regularly convene workshops and symposia that bring communities together around shared problems.

Appointments and responsibilities

- 2026– Editor, special issue "Communication and Causation in Neural Systems", *Philosophical Transactions of the Royal Society B*.
- 2026– Co-organiser, Making Connections Seminar Series.
- 2026– Member of the Neuro-AI MPhil admission committee, MRC-CBU.
- 2025– Coordinator, Cambridge Networks Network.
- 2025– Founding member, Bernstein society node in Hamburg.
- 2025– Reviewing-board member, CALM dataset, MRC-CBU.
- 2022–24 PhD student representative, University of Hamburg.
- 2017–18 Student representative on the MSc admissions committee, University of Oldenburg.
- 2016–18 Founding member of the MSc Neuroscience student body, University of Oldenburg.

Organized events

- 2026 Symposium, "From Principles of Brain Connectivity to Neuro-inspired AI", Hamburg, Germany.
- 2026 Workshop, "Promises & Perils of Connectome-Constrained Neuro-AI Models", OHBM, Bordeaux, France.
- 2025 Workshop on Visual Science Communication, Robinson College, Cambridge, UK.

- 2025 Workshop, "Linking Structure, Dynamics, & Function in Neuronal Networks: Old Challenges & New Directions", OCNS, Florence, Italy.
- 2024 Seminar series, "Networks in Artificial Neural Networks", ICNS-UKE.
- 2023 Symposium, "Modeling Large-scale Brain Networks & Their Disconnection in Neurological Disorders", DGKN, Hamburg, Germany.
- 2022 Symposium, "Searching for Causes & Mechanisms in the Brain", ICNS-UKE.
- 2022 Workshop on Visual Science Communication, Interdisciplinary College, Möhnesee-Günne, Germany.
- 2018 Workshop on Visual Science Communication, University of Oldenburg, Germany.

Peer reviewing experience

I have reviewed for both journals including *Nature Communications*, *PNAS*, *Brain*, *Brain Communications*, *PLOS Computational Biology*, *Network Neuroscience*, and conferences including *IEEE World Congress on Computational Intelligence*, *OHBM Annual Meeting 2024*, and *Cognitive Computational Neuroscience 2024*.

ENGAGEMENT AND BROADER IMPACT

From 2012, I have worked as a graphic designer (part-time along my studies) and I have carried that skillset into my academic career. I communicate complex methods through digestible figures and run workshops on visual science communication. I contribute design to communities, conferences, and institutes, e.g., designing brand material for Cambridge's Mind and Brain Institute and Neuromatch Academy. I treat clear visual communication not as mere decoration but as part of doing accessible science. After all, if a tree falls in a forest and no one is around to hear it, does it make a sound?

- 2022– Graphic designer and member of the volunteer-engagement committee, Neuromatch Academy.
- 2022–25 Graphic designer, Interdisciplinary College.
- 2014–16 Graphic designer, Eshareh creative solutions.
- 2009–16 Graphic designer, Behnam Daheshpour charity organisation for cancer patients.
- 2013–15 Graphic designer, United Nations World Food Programme (WFP).

EDUCATION

- 2019 – 2024: **PhD** (*summa cum laude*), ICNS-UKE.
Thesis: A Game-theoretical Framework for Causal Inference, With Applications in Artificial Neural Networks and Large-scale Models of Brain Dynamics.
- 2016 – 2018: **MSc** Neuroscience, University of Oldenburg, Germany.
- 2018 Research assistant, Montreal Neurological Institute, McGill University, Canada.
- 2017 Research assistant, Timone Institute, Marseille University, France.
Dissertation: Mapping the cortical processing of attended and unattended speech during selective listening.

AWARDS AND SCHOLARSHIPS

2026	Best Poster Award, Cambridge Neuroscience Seminar.	2018	DFG research unit 1735 travel award.
2025	Bernstein travel award.	2018	DAAD scholarship for international students (€1,400).
2022	Best PhD Paper Award, University of Hamburg (€1,000).	2018	University of Oldenburg travel award.
2021	Best PhD Poster Award, University of Hamburg.	2017	A*Midex fellowship, University of Marseille (€6,000).
2018	Volkswagen Foundation SMARTstart fellowship (€30,000).		

INVITED TALKS

2026	<i>From Principles of Brain Connectivity to Neuro-inspired AI</i> , University of Hamburg, Germany.	2026	Workshop on Lesion-symptom Mapping, UKE, Germany.
2026	Organization for Human Brain Mapping (OHBM) Annual Meeting, Bordeaux, France.	2026	Frontiers and Foundations Workshop, University of Cambridge, UK.
2026	Rutgers Center for Advanced Human Brain Imaging Research (CAHBIR) Network Neuroscience Seminar Series.*	2026	Neuroimaging Methods day, MRC-CBU, UK.
2026	Departmental talk, Max Planck Institute for Human Cognitive and Brain Sciences. Leipzig, Germany.*	2025	Bernstein Conference, Frankfurt, Germany.
2026	Lab talk, Department of Psychiatry, University of Cambridge, UK.	2024	Career Day, University of Oldenburg, Germany.
2026	<i>Toward a More Ecological Investigation of Bilingualism: Current Challenges and (AI-based) Solutions</i> , University of Cambridge, UK.	2023	Hamburg Institute for Advanced Study, Germany.
		2023	Hamburg Center for Neuroscience, Germany.
		2022	Brain Modes Conference, Viña del Mar, Chile.*
		2022	SNUFA seminar series, London, UK.*
		2022	IEEE World Congress on Computational Intelligence, Padova, Italy.

*held online

OPEN-SOURCE SOFTWARE

MSA: Multiperturbation Shapley value Analysis: a compact Python package for game-theoretic causal attribution (with S. Dixit, 2021).

YANAT: Yet Another Network Analysis Toolkit (with S. Dixit, 2024).

Hard-working Memory: a unified working- and short-term-memory task for artificial neural networks (2020).

SELECTED PUBLICATIONS

- 1 **Fakhar K**, Astle DE. [Embracing the suboptimal organization of the human brain](#). *Trends in Cognitive Sciences*. 2026.
- 2 **Fakhar K**, Akarca D, Luppi AI, Oldham S, Hadaeghi F, Vértes PE, Bullmore E, Hilgetag C, Astle D. [Human cortical networks trade communication efficiency for computational reliability](#). [In press in *Science Advances*]
- 3 Poli F, **Fakhar K**, Mousley A, Klein C, Grosse Weismann C, Astle D. [Curiosity shapes brain-like architectures and functions](#). 2026. [Under review in *Cell*]
- 4 Jelen M, Mousley A, **Fakhar K**, Trachtenberg E, He Y, Kohler R, Aggarwal S, Warrier V, Bzdok D, Yip S, Astle D. [The topology of adolescent mental health](#). 2026. [Under review in *Nature Communications*]
- 5 Hadaeghi F, **Fakhar K**, Hilgetag CC. [Controlling reciprocity in binary and weighted networks: A novel density-conserving approach](#). *Chaos: An Interdisciplinary Journal of Nonlinear Science*. 2026.
- 6 Hadaeghi F, **Fakhar K**, Khajehnejad M, Hilgetag CC. [A computational perspective on the no-strong-loops principle in brain networks](#). [In press in *Nature Communications*]
- 7 Deng Y, Abdolalizadeh A, **Fakhar K**, Schmitt T, Witt K, Gießing C, Rieger JW, Thiel CM. [Motor- and Cognitive-Dominant Functional Network Adaptations Supporting Dual-Task Performance in Older Adults](#). *Imaging Neuroscience*. 2026.
- 8 **Fakhar K**, Hadaeghi F, Seguin C, Dixit S, Messé A, Zamora-López G, Misic B, Hilgetag CC. [A general framework for characterizing optimal communication in brain networks](#). *eLife*. 2025.
- 9 Dixit S, **Fakhar K**, Hadaeghi F, Mineault P, Kording KP, Hilgetag CC. [Who Does What in Deep Learning? Multidimensional Game-Theoretic Attribution of Function of Neural Units](#). [Under review in *npj Artificial Intelligence*]
- 10 Puschmann S, Regev M, **Fakhar K**, Zatorre RJ, Thiel CM. [Attention-driven modulation of auditory cortex activity during selective listening in a multispeaker setting](#). *Journal of Neuroscience*. 2024.
- 11 **Fakhar K**, Dixit S, Hadaeghi F, Kording KP, Hilgetag CC. [Downstream network transformations dissociate neural activity from causal functional contributions](#). *Scientific Reports*. 2024.
- 12 Moore JA, Wilms M, Gutierrez A, Ismail Z, **Fakhar K**, Hadaeghi F, Hilgetag CC, Forkert ND. [Simulation of neuroplasticity in a CNN-based in-silico model of neurodegeneration of the visual system](#). *Frontiers in Computational Neuroscience*. 2023.
- 13 **Fakhar K**, Hilgetag CC. [Systematic perturbation of an artificial neural network: A step towards quantifying causal contributions in the brain](#). *PLOS Computational Biology*. 2022.
- 14 **Fakhar K**, Hadaeghi F, Hilgetag CC. [Causal influences decouple from their underlying network structure in echo state networks](#). *IEEE-IJCNN*. 2022.
- 15 Popova M, **Fakhar K**, Braun W. [‘One region to control them all’—the surprising effectiveness of network control theory in predicting post-stroke recovery from aphasia](#). *Frontiers in Computational Neuroscience*. 2022.